

developing a special-purpose protocol for sensor networks.

Problems to be addressed

The Bluetooth scatternet formation devices are generally small, running on batteries, with low-powered CPU and low memory. At the same time, these are highly mobile and may frequently move in and out of the network. This leads us to think about:

1. Topology and size of scatternets. What is the optimal number of piconets in the scatternet and how many nodes are there? The number of bridge nodes increases (proportionally) with the increasing number of piconets. The increases, in turn, incur significant overhead: loss of one or more time slots to readjust clocking when shifting from one piconet to another, more traffic than non-bridge nodes and increased battery power consumption to perform these tasks.

What is the optimal number of piconets in which a bridge should participate? The more the piconets with which a bridge communicates, the less the time during which the bridge will be available for dealing with traffic from/to each individual piconet, thus increasing the likelihood of bottlenecks, at the bridge itself.

2. Formation delay and resilience. How much time is required to construct, optimise and maintain the scatternet? The *ad-hoc*, dynamic nature of a Bluetooth network requires constant modifications to the scatternet topology to support devices that join and leave. How to handle the nodes that join or leave the scatternets?

3. Polling and scheduling. In what order will the slaves be polled by the master? Does the scatternet topology impact the ability to handle scheduled communications from slaves in sniff or hold state?

4. Routing. Does the scatternet topology allow efficient routing of messages?

Work in the area

There are a number of papers discussing the effective routing protocols

for Bluetooth scatternets. Also, there are quite a few papers which discuss the point of organising the scatternet in an efficient fashion, dealing with disparate aspects of scatternet optimisation. A proposal in 'Bluetooth Scatternet Formation for Supporting Device Mobility,' by Chong-Hong Yang and Yi-Sheng Chen, deals with mobility support in a Bluetooth scatternet network, but relies on a Bluetooth backbone network consisting of devices that are not mobile. Another work done on similar lines is 'Tracking the Optimal Configuration of a Bluetooth Scatternet' by Csaba Kiss Kalló and Carla-Fabiana Chiasserini, which deals with similar problem of redefining the role of nodes in scatternets but concentrates on improving the energy efficiency of the nodes. Also, there are numerous proposals for routing protocols including MANET (mobile ad-hoc network) protocols.

Future in Bluetooth

The new Bluetooth specification, called the Bluetooth 2.1, is already published. It is fully backward-compatible with 1.1. The specification includes many new security and usability enhancements. The most interesting of these is incorporation of the near-field communication (NFC) device, which helps in pairing of two Bluetooth devices equipped with NFC chip by just bringing them close to each other. For example, a headset can be paired with a Bluetooth 2.1 phone having NFC chip just by bringing the two devices close to each other. Another example is automatic uploading of photos from a mobile phone to a digital picture frame just by bringing the phone close to the frame. Features like this and other security enhancements will help Bluetooth gain user confidence.

A lot of work is in progress in the area of Bluetooth, to enhance and widen the application of Bluetooth protocol. In the future, we may expect features like:

1. Broadcast channels. These will enable Bluetooth information points

that broadcast information to all Bluetooth devices in the vicinity. So users will be able to pull relevant information from the system.

2. Topology manager. It will enable better configuration of piconets and even scatternets.

3. Alternate media access physical layer. It will enable use of some other wireless technology for actual data transfer at higher rate while maintaining the initial pairing on standard Bluetooth.

4. QoS improvements. These will enable audio and video data transmission at a higher quality.

Nokia and Bluetooth SIG have announced that Wibree (Nokia's low power wireless communication protocol) will be a part of the Bluetooth specification as an ultra-low-power Bluetooth technology, to be used in caller ID watches, sports sensors (used for monitoring the wearer's heart rate during exercise) as well as medical devices. A medical devices profile and associated protocols are being developed by the Medical Devices Working Group (MDWG).

Bluetooth 3.0. The next version of Bluetooth after v2.1, code-named Seattle, is proposed to adopt ultra-wideband (UWB) radio technology. This will allow Bluetooth use over UWB radio, enabling very fast data transfers of up to 480 Mbps, while taking the advantage of very low-power idle modes of Bluetooth.

Extending the scope

Even though work on Bluetooth scatternet is going on for about a decade now, it has failed to generate interest amongst general consumer electronics goods manufacturers. Bluetooth scatternets find most of their use in sensor networks. When there is a limitation on setting up the communication infrastructure, Bluetooth scatternet could turn out to be a cost-effective solution. In future, as the processing power of electronic devices grows and they become more energy-efficient, Bluetooth might see its way into already evolving Bluetooth standards. ●